

```
-- SECTION 09 - TRIGGERS
-- Easy and Medium Tasks Only
-- TASK 1: BEFORE INSERT - Auto Assign emp_ID

SET SERVEROUTPUT ON;

-- Create EMP_SEQ if it does not already exist
BEGIN
    EXECUTE IMMEDIATE '
        CREATE SEQUENCE EMP_SEQ
        START WITH 1000
        INCREMENT BY 1
    ';

    DBMS_OUTPUT.PUT_LINE('EMP_SEQ created.');
```

Script Output X

Task completed in 0.127 seconds

PL/SQL procedure successfully completed.

Trigger TRG_EMP_ID compiled

Trigger TRG_EMP_ID compiled

```
VALUES (
    'Trigger',
    'Test',
    'T',
    21,
    'Muscat',
    'trigger.task1@test.com',
    'pass123',
    1,
    1
);

-- Verify emp_ID was auto assigned
```

Script Output X

Task completed in 0.092 seconds

Trigger TRG_EMP_ID compiled

Savepoint created.

1 row inserted.

EMP_ID	NAME	EMP_EMAIL
1000	Trigger	trigger.task1@test.com

Rollback complete.

TEST_EMPLOYER_COUNT
0

emp_ID = 1000

```

RAISE;
END IF;
END;

-- Create trigger to insert welcome log after new employee insert
CREATE OR REPLACE TRIGGER TRG_EMP_WELCOME_LOG
AFTER INSERT ON EMPLOYEE
FOR EACH ROW
BEGIN
    INSERT INTO EMPLOYEE_LOG (
        log_ID,
        emp_ID,
        action,
        log_timestamp
    )
    VALUES (
        EMPLOYEE_LOG_SEQ.NEXTVAL,
        :NEW.emp_ID,
        'NEW HIRE',
        SYSDATE
    );
END;

```

Script Output .x | Task completed in 0.11 seconds

TEST_EMPLOYEE_COUNT

0

EMPLOYEE_LOG table created.

PL/SQL procedure successfully completed.

EMPLOYEE_LOG_SEQ created.

PL/SQL procedure successfully completed.

Trigger TRG_EMP_WELCOME_LOG compiled

Trigger TRG_EMP_WELCOME_LOG compiled

1. TRG_EMP_emp_ID
2. TRG_EMP_WELCOME_LOG EMPLOYEE_LOG NEW HIRE

```

-- Verify log was inserted automatically
SELECT l.log_ID,
       l.emp_ID,
       l.action,
       l.log_timestamp
FROM EMPLOYEE_LOG l
WHERE l.emp_ID = e.emp_ID
       OR l.emp_ID = e.emp_ID
       AND e.emp_email = 'welcome.logtest@test.com';

-- Rollback test data
ROLLBACK TO before_trg_welcome_log_test;

-- Final check: employee should be removed
SELECT COUNT(*) AS test_employee_count
FROM EMPLOYEE;

```

Script Output .x | Task completed in 0.142 seconds

1 row inserted.

EMP_ID	NAME	EMP_EMAIL
1001	Welcome	LogTest

LOG_ID	EMP_ID ACTION	LOG_TIMES
1	1001 NEW HIRE	16-JUN-26

Rollback complete.

TEST_EMPLOYEE_COUNT

0

TEST_LOG_COUNT

1

EMP_ID = 1001

ACTION = NEW HIRE

Rollback complete

TEST_EMPLOYEE_COUNT = 0

Worksheet: Query Builder

```

-- LogTest,
-- 12,
-- 10,
-- 'Mascot',
-- 'welcome_two@test.com',
-- 'pass123',
-- 1,
-- 2
--
-- Verify that employees were inserted
SELECT emp_id,
       last_name,
       emp_email
FROM EMPLOYEES
WHERE emp_email IN ('welcome_one@test.com', 'welcome_two@test.com')

```

Script Output: X | Task completed in 0.114 seconds

TEST_LOG_COUNT

0

Severpoint created.

1 row inserted.

1 row inserted.

EMP_ID	LAST_NAME	EMP_EMAIL
1002	WelcomeOne	welcome_one@test.com
1003	WelcomeTwo	welcome_two@test.com

LOG_ID	EMP_ID	LAST_NAME	ACTION	LOG_TIMES
2	1002	WelcomeOne	NEW HIRE	14-JUN-16
3	1003	WelcomeTwo	NEW HIRE	14-JUN-16

Rollback complete.

Activate Windows
Go to Settings to activate Windows.

Salary decrease is not allowed. /Trigger TRG_PREVENT_SALARY_CUT compiled

```

-- TASK 3: BEFORE UPDATE - Prevent Salary Decrease
--
--
SET SERVEROUTPUT ON;
CREATE OR REPLACE TRIGGER TRG_PREVENT_SALARY_CUT
BEFORE UPDATE OF amount ON SALARY_BONUS
FOR EACH ROW
BEGIN
    IF :NEW.amount < :OLD.amount THEN
        RAISE_APPLICATION_ERROR(-20001, 'Salary decrease is not allowed.');

```

Script Output: X | Task completed in 0.054 seconds

Rollback complete.

TEST_EMPLOYEE_COUNT

0

Trigger TRG_PREVENT_SALARY_CUT compiled

```

0

Trigger TRG_PREVENT_SALARY_CUT compiled

Savepoint created.

SALARY_ID  JOB_ID  AMOUNT  ANNUAL  BONUS
-----
3          3      4500    54000    600

Expected salary decrease error: ORA-20001: Salary decrease is not allowed.
ORA-06512: at "KAWTHER_EMS.TRG_PREVENT_SALARY_CUT", line 3
ORA-04088: error during execution of trigger 'KAWTHER_EMS.TRG_PREVENT_SALARY_CUT'

PL/SQL procedure successfully completed.

SALARY_ID  JOB_ID  AMOUNT  ANNUAL  BONUS
-----
3          3      4500    54000    600

1 row updated.

SALARY_ID  JOB_ID  AMOUNT  ANNUAL  BONUS
-----
3          3      4600    54000    600

Rollback complete.

SALARY_ID  JOB_ID  AMOUNT  ANNUAL  BONUS
-----
3          3      4500    54000    600

```

Expected salary decrease error: ORA-20001

Salary decrease is not allowed.

AMOUNT = 4600

AMOUNT = 4500

```

Task completed in 0.071 seconds

SALARY_ID  JOB_ID  AMOUNT  ANNUAL  BONUS
-----
3          3      4500    54000    600

Expected salary decrease error: ORA-20001: Salary decrease is not allowed.
ORA-06512: at "KAWTHER_EMS.TRG_PREVENT_SALARY_CUT", line 3
ORA-04088: error during execution of trigger 'KAWTHER_EMS.TRG_PREVENT_SALARY_CUT'

PL/SQL procedure successfully completed.

SALARY_ID  JOB_ID  AMOUNT  ANNUAL  BONUS
-----
3          3      4500    54000    600

1 row updated.

SALARY_ID  JOB_ID  AMOUNT  ANNUAL  BONUS
-----
3          3      4600    54000    600

Rollback complete.

SALARY_ID  JOB_ID  AMOUNT  ANNUAL  BONUS
-----
3          3      4500    54000    600

EMPLOYEE_ARCHIVE table created.

PL/SQL procedure successfully completed.

Trigger TRG_ARCHIVE_EMPLOYEE compiled

```

Activate Wii
Go to Settings t

EMPLOYEE_ARCHIVE table created. /Trigger TRG_ARCHIVE_EMPLOYEE compiled

```
        lname,
        emp_email
FROM EMPLOYEE
WHERE emp_email = 'archive.test@test.com';

-- Delete employee to activate archive trigger
DELETE FROM EMPLOYEE
WHERE emp_email = 'archive.test@test.com';

-- Verify employee was deleted
SELECT COUNT(*) AS employee_count_after_delete
FROM EMPLOYEE
WHERE emp_email = 'archive.test@test.com';

-- Verify employee was archived automatically
SELECT emp_id,
       fname,
```

Script Output

Task completed in 0.115 seconds

EMPLOYEE_ID	EMPLOYEE_NAME	EMPLOYEE_EMAIL
1004	Archive	Test
archive.test@test.com		

1 row deleted.

EMPLOYEE_COUNT_AFTER_DELETE
0

EMP_ID	FNAME	LASTNAME	EMP_EMAIL
1004	Archive	Test	archive.test@test.com

Rollback complete.

ARCHIVE_COUNT_AFTER_ROLLBACK
0

Activate Wi-Fi
Go to Settings

1 row deleted.

EMPLOYEE_COUNT_AFTER_DELETE = 0

Archive Test

archive.test@test.com

ARCHIVE_COUNT_AFTER_ROLLBACK = 0